# Dimension Sheet: Mars Rover Corner Wheel & Drive Module

ME-109 Engineering Drawing Project. **All members model from this sheet.** All dimensions are in mm unless stated otherwise. Rev A, 11 Nov 2026.

## 0. How to Read This Sheet

Every dimension has a **Source** code that explains where the number comes from:

| Code | Meaning |
|---|---|
| **REF** | Taken from a published source about the real Curiosity rover (reference list in Section 14) |
| **PHOTO** | Estimated by scaling a NASA photo against the known 500 mm wheel (method in Section 13) |
| **STD** | Fixed by a standard or a catalogue part (bearing, key, bolt, fit table) |
| **RULE** | A design rule of thumb from machine-design textbooks |
| **FIT** | Derived from a mating part, so that the two parts fit (the related part is named) |
| **CHOICE** | A design decision made by our group (the reason is given). Every real design has some of these. |

**Global coordinate system (for assembly):** the origin is the centre of the wheel. **X** runs along the drive axle, with **+X pointing inboard** (toward the rover body). **Z** points up and is the steering axis. **Y** completes the right-hand system. Every part's position below is given in these coordinates, so the assembly owner can place each part with a single move.

**Untoleranced dimensions:** ISO 2768-m (medium) general tolerances **[R8]**. Write this in every title block.
**Default edges:** external corners have a 1 × 45° chamfer and internal corners have an R1 fillet, unless stated otherwise **[RULE, R10]**.

## 1. Overall Layout (Axial Stack along X)

| Item | X from → to | Why it sits here |
|---|---|---|
| Axle nut + washer | −238 → −222 | Clamps the hub outboard |
| 03 Drive hub (flange / spigot / boss) | −222 → −160 | Outboard, so the wheel can be removed without opening the motor |
| 02 Spoke plate (inner ring) | −210 → −200 | Sandwiched between the hub flange and the rim ring |
| 01 Wheel rim | −200 → +200 | 400 mm wide, centred on the origin [REF R1] |
| 05 Motor housing | −150 → +60 | Fits inside the wheel (an in-hub motor, as on Curiosity [PHOTO R2]) |
| Bearing B1 (6007) | −146 → −132 | In the housing end wall |
| 09 Bearing spacer | −132 → +34 | Fills the gap between the two bearings |
| Bearing B2 (6007) | +34 → +48 | In the end-cap boss |
| 04 Axle | −242 → +51 | From the thread end to just past the circlip |
| 06 Motor end cap | +34 → +72 | Closes the housing on the inboard side |
| 07 Steering bracket | X −45 → +255, Z −20 → +336 | C-shape that wraps around the inboard side and the top of the wheel |
| Bearing B3 (6007) | Z +322 → +336 | Steering bearing, on the steering axis |
| 08 Steering actuator housing | Z +321 → +471 | Above the wheel, as in Fig. 1 of the proposal [PHOTO R2] |

Bearing arrangement: the inner rings of B1 and B2 are clamped between the axle shoulder (X −146) and a circlip (X +48). The outer rings are *cross-located*: B1 against the housing lip and B2 against the end-cap shoulder. This is a standard

textbook arrangement for a short shaft [R4, R10].

## 2. Part 01: Wheel Rim (Al 7075-T6)

Position: X −200 → +200, coaxial with X.

| Feature | Dimension | Source | How we got it |
|---|---|---|---|
| Outer diameter of skin | Ø500 | REF | Curiosity's wheel is about 50 cm in diameter [R1] |
| Width | 400 | REF | Curiosity's wheel is about 40 cm wide [R1] |
| Skin thickness | 3 (ID Ø494) | CHOICE | The real skin is about 0.75 mm [R3]. We thickened it to 3 mm so it is visible and dimensionable on an A4 sheet at 1:5. |
| Outboard stiffening ring | Ø494 → Ø460, 8 thick (X −200 → −192) | CHOICE / PHOTO | The rim needs a ring for the spokes to bolt to [PHOTO R2: PIA13237]. 17 mm of radial depth gives room for M5 holes. |
| Spoke mounting holes | 6 × M5 tapped through, PCD 478, at 0°, 60° … 300° | STD / FIT | One hole per spoke (Part 02). An M5 thread needs 1.5 × d = 7.5 mm of engagement in aluminium [RULE R10]; the ring is 8 thick. Tap drill Ø4.2 [R7]. |
| Inboard lip | Ø494 → Ø484, 5 thick (X +195 → +200) | RULE | Stiffens the open edge of a thin shell |
| Grousers (treads) | 24 off, at 15° spacing | PHOTO / CHOICE | Count them in PIA16134 [R2]. 24 gives a clean 15° polar array. |
| Grouser height × thickness | 7 × 4 | PHOTO | Scaled from PIA20334 (the grousers are about 1/70 of the wheel diameter) |
| Grouser shape | Chevron (V): apex at X = 0, arm ends at X = ±196, ends offset 6° behind the apex | PHOTO | Matches the zig-zag pattern in Fig. 1. 6° on Ø500 is about 26 mm. |
| Overall diameter over grousers | Ø514 | FIT | 500 + 2 × 7 |

**How to model it:** REVOLVE the half-section (skin, ring and lip). Model each grouser arm as a 4 × 7 BOX that sits 1 mm into the skin, rotated by 3°. MIRROR it to make the V, ARRAYPOLAR 24 copies, then UNION everything.

## 3. Part 02: Spoke Plate (Ti-6Al-4V)

Position: X −210 → −200. It lies flat in the YZ plane.

| Feature | Dimension | Source | How we got it |
|---|---|---|---|
| Inner ring | OD Ø150, ID **Ø80 H7** (+0.030/0), 10 thick | FIT / STD | The ID pilots on the hub spigot Ø80 g6 to give a clearance locating fit [R5]. The OD matches the hub flange. |
| Hub bolt holes | 6 × M8 tapped through, PCD 120, at 30°, 90° … 330° | FIT / RULE | Match the hub flange holes. Offset 30° from the spokes so the holes do not cut the spoke roots. Thread engagement of 1 × d is enough in titanium [R10]. |
| Number of spokes | 6, at 0°, 60° … 300° | PHOTO | Count them in PIA13237 [R2] |
| Spoke section | 8 (in-plane) × 10 | CHOICE | Thin in the plane so the spoke can flex like |

| | | | |
|---|---|---|---|
| | (thickness along X) | | a spring, which is the job of the real spokes [R1, R3] |
| Spoke path (spline fit points, Y,Z for the 0° spoke) | (72,0) → (120,25) → (170,−10) → (232,0) | PHOTO / CHOICE | Gives an S-curve that resembles PIA13237 |
| Spoke end tab | 16 (radial) × 20, 10 thick, Ø5.5 hole at r = 239 | STD / FIT | Clearance hole for M5 is Ø5.5 (ISO 273, medium) [R7]. It lines up with the rim's PCD 478. |
| Root fillets | R5 | RULE | Reduces stress concentration where the spoke meets the ring |

**How to model it:** draw the spline and SWEEP an 8 × 10 rectangle along it. Add the tab, ARRAYPOLAR 6 copies, then UNION with the inner ring.

## 4. Part 03: Drive Hub (Al 6061-T6)

Position: X −222 → −160.

| Feature | Dimension | Source | How we got it |
|---|---|---|---|
| Bore | **Ø30 H7** (+0.021/0), through | STD / FIT | Transition fit with the axle's Ø30 k6, which is usual for a keyed hub [R5, R10] |
| Flange | Ø150 × 12 (X −222 → −210) | RULE | Flange OD is about 5 × bore. A thickness of 12 is 1.5 × the M8 bolt diameter. |
| Flange holes | 6 × Ø9 through, PCD 120, at 30°, 90° … | STD | Clearance hole for M8 is Ø9 (ISO 273, medium) [R7]. Edge distance (75 − 60 − 4.5 = 10.5) is more than 1 × d [RULE]. |
| Spigot | **Ø80 g6** (−0.010/−0.029) × 10 (X −210 → −200) | FIT | Locates the spoke plate (Ø80 H7) concentric with the axle |
| Boss | Ø60 × 40 (X −200 → −160) | RULE | Boss OD is about 2 × bore [R10]. The 62 mm total length is about 2 × bore, which gives good key engagement. |
| Keyway (hub) | 8 **JS9** (±0.018) wide, depth t2 = 3.3 $^{+0.2}$ | STD | Shafts over 22 and up to 30 mm take an 8 × 7 key (DIN 6885-1 / ISO 773) [R6] |
| Edges | C1 on the bore and flange, R2 between the flange and the boss | RULE | |

## 5. Part 04: Axle (Steel AISI 4340)

Position: X −242 → +51. Segments are listed from outboard to inboard.

| Segment / feature | Dimension (X range) | Source | How we got it |
|---|---|---|---|
| Thread | M24 × 2, length 20 (−242 → −222) | STD / FIT | Smaller than the Ø30 hub seat, so the hub slides over it. A thin nut (ISO 8675, m = 12) plus a washer (ISO 7089, 4 thick) uses 16 mm, leaving 4 mm of protrusion [R7]. |
| Hub seat | **Ø30 k6** (+0.015/+0.002), length 62 (−222 → | FIT / STD | Same length as the hub [R5] |

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|---|---|---|---|
| | −160) | | |
| Keyway (shaft) | 8 **N9** (0/−0.036) wide, depth t1 = 4.0 $^{+0.2}$, length 50 (−218 → −168), round ends | STD | DIN 6885-1 for an 8 × 7 key [R6]. Key length 50 is less than the hub length of 62. |
| Shoulder | Ø40, length 14 (−160 → −146) | STD / FIT | Locates the hub on one face and bearing B1 on the other. Ø40 is within the SKF abutment diameter for a 6007 ($d_a$ min = 40) [R4]. It also passes through the Ø45 hole in the housing with clearance. |
| Bearing / spacer seat | **Ø35 k6** (+0.018/+0.002), length 194 (−146 → +48) | STD | SKF recommends k-fits for a rotating inner ring under normal load, d 18–100 mm [R4]. One diameter all the way along keeps it easy to grind. |
| Circlip groove | Ø33.0 × 1.6 wide (+48 → +49.6) | STD | DIN 471 retaining ring for a 35 mm shaft (d2 = 33, m = 1.6) [R7] |
| End | Ø35, to X +51, C1 | RULE | Leaves material beyond the groove |
| Shoulder fillets | R0.6 | STD | Must be smaller than the 6007's ring chamfer (r = 1.0) so the bearing seats flat [R4] |
| Thread end | C1.5 chamfer, thread relief 3 wide | RULE | Show the thread with the cosmetic convention (thin line at the root diameter) |

Total length is 293. On the drawing, show it horizontal at 1:2 with a broken view or a revolved section through the keyway.

## 6. Part 05: Motor Housing (Al 6061-T6)

Position: X −150 → +60.

| Feature | Dimension | Source | How we got it |
|---|---|---|---|
| Body | OD Ø150, ID **Ø136 H7** (+0.040/0) | PHOTO / RULE | The hub motor is about 0.3 × the wheel diameter in PIA13236 [R2]. The 7 mm wall is in the usual range for cast or machined aluminium housings [R10]. H7 because the end-cap spigot fits into this bore. |
| Outboard end wall | 18 thick (−150 → −132) | FIT | Holds the 14-wide bearing plus a 4 mm lip |
| Bearing bore B1 | **Ø62 H7** (+0.030/0), depth 14 (−146 → −132) | STD | The 6007 has a 62 OD and is 14 wide [R4]. SKF recommends H7 for a stationary outer ring [R4]. |
| Axle hole | Ø45 through the lip (−150 → −146) | STD / FIT | Smaller than the SKF housing abutment $D_a$ max, so the lip holds the outer ring; larger than the Ø40 shoulder for clearance |
| Mounting flange | Ø210 × 12 (+48 → +60) | RULE | Thickness 1.5 × M8 = 12 for thread engagement in aluminium [R10] |
| Flange threads | 6 × M8 tapped through, PCD 184, at 0°, 60° … | STD / FIT | Tap drill Ø6.8 [R7]. The holes sit at r 92 ± 4, which is outside the Ø150 body. Matches the end cap and bracket. |
| Cooling fins | 12 off at 30°, 3 thick, 8 high (to Ø166), X | CHOICE | Adds a polar array and modeling variety. Clears the hub boss and the rim. |

| | | | |
|---|---|---|---|
| | −130 → +40 | | |
| Fillets | R3 internal corners, R1 fin roots | RULE | |

## 7. Part 06: Motor End Cap (Al 6061-T6)

Position: X +34 → +72.

| Feature | Dimension | Source | How we got it |
|---|---|---|---|
| Flange | Ø210 × 12 (+60 → +72) | FIT | Same as the housing flange |
| Flange holes | 6 × Ø9 through, PCD 184 | STD | Clearance for M8 [R7] |
| Spigot | **Ø136 g6** (−0.014/−0.039) × 10 (+50 → +60) | FIT / STD | Clearance locating fit into the housing's Ø136 H7 [R5] |
| O-ring groove (on the spigot) | 4.0 wide × 2.3 deep, centred at X +55 | STD | Groove for a 3 mm cross-section O-ring, depth about 75–80% of the cross-section [R9]. Keeps dust out. |
| Bearing boss | OD Ø80 (+34 → +50) | RULE | Leaves a 9 mm wall around the bearing |
| Bearing bore B2 | **Ø62 H7**, depth 14 (+34 → +48) | STD | Same as B1 [R4] |
| Relief pocket | Ø50, depth 5 (+48 → +53) | FIT | Clearance for the axle end and the circlip; the shoulder at +48 holds B2's outer ring |
| Cable port | Ø20 through, at r = 60, angle 90° | CHOICE | Passes the motor wiring. Positioned clear of the bolt circle and the boss. |

## 8. Part 07: Steering Bracket (Al 7075-T6)

A C-shaped bar, 60 wide in Y (Y −30 → +30).

| Feature | Dimension | Source | How we got it |
|---|---|---|---|
| Mounting pad | Ø210 disc × 15 (X +72 → +87), centred on the drive axis | FIT | Bolts to the end cap and housing flange |
| Pad holes | 6 × Ø9, PCD 184, plus a Ø24 cable hole at r = 60, 90° | STD / FIT | M8 clearance. The cable hole lines up with the end-cap port. |
| Lower arm | X +87 → +230, Z −20 → +20 | CHOICE | Leaves the wheel through its open inboard side |
| Vertical leg | X +230 → +255, Z −20 → +305 | FIT | 30 mm clear of the rim edge (X 200). Total height clears the grouser tips (Z 257). |
| Upper arm | X −45 → +255, Z +280 → +305 | FIT | 23 mm clear above the grouser tips (Z 257). Reaches the steering axis at X = 0. |
| Steering boss | Ø90, Z +280 → +320, at X 0, Y 0 | RULE | Boss OD is about 2.5 × the bearing bore |
| Steering spigot | Shoulder Ø42 (Z 320 → 322), then **Ø35 k6** (Z 322 → | STD | Carries bearing B3 (6007) [R4]. The shoulder stops the housing from rubbing on the boss face. |

| | | | |
|---|---|---|---|
| | 336) | | |
| Corner gussets | Triangles with 50 mm legs, 20 thick, at both inside corners | RULE | Stiffen the C. They also give the drawing an extra section to show. |
| Fillets | R10 inside corners, R3 outside | RULE | |

**How to model it:** draw the C profile with PLINE in the XZ plane and EXTRUDE it 60. UNION the pad, boss, spigot and gussets. Cut the holes with SUBTRACT.

## 9. Part 08: Steering Actuator Housing (Al 6061-T6)

Position: on the Z axis, Z +321 → +471.

| Feature | Dimension | Source | How we got it |
|---|---|---|---|
| Body | OD Ø120 × 150 tall | PHOTO | About 0.25 × the wheel diameter in PIA16134 [R2] |
| Bottom wall | 19 thick (Z 321 → 340) | FIT | Holds the bearing (14) plus a lip |
| Bearing bore B3 | **Ø62 H7**, depth 15 (Z 321 → 336) | STD | For the 6007 [R4]. The extra 1 mm leaves a gap above the bracket shoulder. |
| Lip hole | Ø45 (Z 336 → 340) | FIT | Same approach as the motor housing |
| Internal cavity | Ø100, Z 340 → 459 | RULE | 10 mm wall. This space holds the steering motor (not modeled). |
| Top flange | Ø170 × 12 (Z 459 → 471) | RULE | Mounts to the rover's suspension arm |
| Flange holes | 6 × Ø9, PCD 150 | STD | M8 clearance [R7] |

## 10. Part 09: Bearing Spacer (Steel)

| Feature | Dimension | Source | How we got it |
|---|---|---|---|
| Length | **166 ± 0.05** (X −132 → +34) | FIT | Exact distance between the two bearing inner rings. Tight tolerance because it sets the bearing spacing. |
| Bore | Ø35.5 | FIT | Slides over the Ø35 k6 seat with clearance |
| Outside diameter | Ø42 | STD | Must press only on the inner ring: within the 6007 inner-ring abutment range ($d_a$ about 40–43) [R4] |
| End faces | Parallel within 0.02 (geometric tolerance) | RULE | Gives a simple example of a geometric tolerance on the drawing |
| Edges | C0.5 | RULE | |

## 11. Standard Parts (Bill of Materials)

| Item | Part | Specification | Qty | Source |
|---|---|---|---|---|
| 10 | Deep-groove ball bearing | SKF 6007: 35 × 62 × 14, r = 1.0 | 3 | [R4] |
| 11 | Parallel key | 8 × 7 × 50, form A, DIN 6885-1 | 1 | [R6] |
| 12 | Thin hex nut | M24 × 2, ISO 8675 | 1 | [R7] |
| 13 | Plain washer | ISO 7089, size 24 (25 × 44 × 4) | 1 | [R7] |

| | | | | |
|---|---|---|---|---|
| 14 | Retaining ring (circlip) | DIN 471, 35 × 1.5 | 1 | [R7] |
| 15 | Socket head cap screw | M8 × 20, ISO 4762 (hub to spoke plate) | 6 | [R7] |
| 16 | Socket head cap screw | M8 × 35, ISO 4762 (bracket + end cap → housing) | 6 | [R7] |
| 17 | Socket head cap screw | M5 × 16, ISO 4762 (spoke tabs → rim) | 6 | [R7] |
| 18 | O-ring | Ø128 ID × 3 cross-section, NBR | 1 | [R9] |

Bolt length check:

- **M8 × 35:** bracket pad (15) + end-cap flange (12) leaves 8 mm of thread engaged in the housing flange.
- **M8 × 20:** hub flange (12) leaves 8 mm engaged in the spoke plate.
- **M5 × 16:** spoke tab (10) leaves 6 mm engaged in the rim ring.

You can draw the standard parts as simplified shapes. They appear in the BOM and the assembly but do not need their own sheets.

## 12. Summary of Fits and Tolerances (for the drawings)

| Interface | Fit | Limits | Type |
|---|---|---|---|
| Bearing on axle / bracket spigot | Ø35 k6 | +0.018 / +0.002 | Light interference |
| Bearing in housing / cap / steering housing | Ø62 H7 | +0.030 / 0 | Locating (sliding) |
| Hub on axle | Ø30 H7/k6 | H +0.021/0; k +0.015/+0.002 | Transition |
| Spoke plate on hub spigot | Ø80 H7/g6 | H +0.030/0; g −0.010/−0.029 | Clearance (locating) |
| End cap in housing | Ø136 H7/g6 | H +0.040/0; g −0.014/−0.039 | Clearance (locating) |
| Key in shaft / hub | 8 N9 / 8 JS9 | 0/−0.036; ±0.018 | Normal key fit |
| Spacer length | 166 ± 0.05 | | Functional length |

All limit values come from ISO 286-2 tables [R5]. Any engineering-drawing or machine-design textbook appendix gives the same numbers, so check them there.

## 13. How to Estimate a Size from a Photo (PHOTO method)

1. Open the NASA photo in any image viewer that shows pixel coordinates (GIMP, Paint, or Photopea online).
2. Measure the wheel's diameter in pixels across a straight-on part of the image, for example 400 px.
3. Scale factor = 500 mm ÷ 400 px = 1.25 mm per px.
4. Measure the feature in pixels and multiply by the scale factor. Use only features roughly in the same plane as the wheel, because perspective distorts the rest.
5. Round to a sensible number and write it on this sheet as PHOTO. This typically gets within 10–20% of the real size, which is accurate enough for a design inspired by the rover.

## 14. References


1. **[R1]** The Planetary Society, "Diagram of a Curiosity wheel." https://www.planetary.org/space-images/diagram-of-a-curiosity-wheel. Gives the wheel diameter (about 50 cm), width (about 40 cm) and spoke layout.
2. **[R2]** NASA/JPL-Caltech Photojournal images PIA16134, PIA20334, PIA13235, PIA13236 and PIA13237. https://photojournal.jpl.nasa.gov. Used for proportions (PHOTO).
3. **[R3]** NASA/JPL news releases on Curiosity wheel wear (2014 onwards). Describe the thin aluminium skin (about

0.75 mm) and the flexible titanium spokes.

4. **[R4]** SKF, *Rolling Bearings* catalogue / SKF product page for bearing 6007 (skf.com). Gives the dimensions (35 × 62 × 14), the abutment diameters $d_a$ and $D_a$, and the recommended shaft and housing fits.
5. **[R5]** ISO 286-2, *ISO system of limits and fits: tables of standard tolerance classes*. The H7, g6 and k6 values are reproduced in the appendix of N.D. Bhatt's *Machine Drawing*.
6. **[R6]** DIN 6885-1 / ISO 773, *Parallel keys and keyways*. Gives the key size for the shaft diameter and the keyway depths t1 and t2.
7. **[R7]** Fastener standards:
    - ISO 261: metric threads
    - ISO 273: clearance holes
    - ISO 4762: socket head cap screws
    - ISO 7089: washers
    - ISO 8675: thin nuts
    - DIN 471: shaft circlips

Tap-drill sizes come from any thread chart (M5 → 4.2, M8 → 6.8).

8. **[R8]** ISO 2768-1, *General tolerances*. Medium class ("m") for untoleranced dimensions.
9. **[R9]** Parker Hannifin, *O-Ring Handbook* (ORD 5700). Gives the groove width and depth for a given O-ring cross-section.
10. **[R10]** N.D. Bhatt, *Machine Drawing*, and R.G. Budynas & J.K. Nisbett, *Shigley's Mechanical Engineering Design*. Source of the proportion rules (boss ≈ 2 × bore, thread engagement 1–1.5 × d, wall thicknesses, fillets) and the bearing-arrangement conventions.

**Viva tip:** for every dimension on your own sheets, be ready to say which category it is (REF, STD, FIT, RULE or CHOICE) and why. That is exactly what the "modelling decisions" part of the viva tests.